Time and Space Complexity

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↳ O → worst

↳ Θ → Avg

↳ ω → Best

$f(x) = x^2 + 2$

O(...)

a = 6 , b = 5;

→ Constant time

O(3) → O(1)

//Swap

↳ int temp = a;

→ a = b;

→ b = temp;

int n = arr.length;

for(int i=0 ; i<n ; i++){

↳ Sout(arr[i]);

}

O(3×n) → O(n)

i = 0

* Points to remember

... values. n → 105

① Always avoid the Constant values. $n \rightarrow 105$

② neglect the lower values.

↳ e.g:- $O(n) + O(5)$

↳ $O(n + \cancel{5})$

↳ $O(n)$

Q: find the time Complexity.

```
i=0  for(int i=0; i<n; i++){
↳n      ↳for(int j=0; j<n; j++){
            Sout(*);
        }
}
```

$n \times n$

$O(n^2)$.

$i=0$, $j=0 \cdots n$

$i=1$, $j=0 \cdots n$

$i=2$, $j=0 \cdots n$

⋮

$i=n-1$ $j=0 \cdots n$

Q: for(int i=0; i<n; i++){ → T.C

↳ for(int j=0; j<i; j++){

$i=n$

$i = n$

sout (*);

}

$n \times n \longrightarrow O(n^2)$

}

$i = 0$ $\qquad$ $j \rightarrow$